# Template Exam Solutions Document

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## Learning objective

Learning Objective Description

## Criteria assessment

This task sheet assesses Criteria CX through CX. Each problem assesses the criterion or criteria indicated in its title, and the number of points available is shown at the end of each item. Partial credit may be awarded when the correct procedure is clearly shown but arithmetic errors are made. Answers that do not include the required procedure will not receive credit. Failure to write your name on the task sheet or submitting it after the deadline will result in a one-point deduction from each criterion.

A minimum of N points is required to pass each criterion.

## Evaluated criteria

**C1** Example criterion description.
**C2** Example criterion description.
**C3** Example criterion description.

## 1 Example Uniform Investment Plan [C1-3 C2-2]

A variable $V$ follows a uniform model on $[8, 16]$. Use the graph and density model to answer the tasks.

⇧

### Questions/Tasks

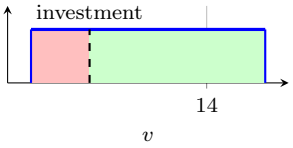
**C1** Write the density function and support. **[3]**
**C2** Compute $P(10 \le V \le 14)$ and justify with area. **[2]**

### C1 - Write the density function and support. [3]

$$f_V(v) = \begin{cases} \dfrac{1}{16-8} = \dfrac{1}{8}, & 8 \le v \le 16, \\ 0, & \text{otherwise.} \end{cases}$$

$$8 \le v \le 16 \quad \longrightarrow \quad f_V(v) = \frac{1}{8} \quad \longrightarrow \quad \int_8^{16} f_V(v)\,dv = 1$$

**C2 - Compute $P(10 \le V \le 14)$ and justify with area.** **[2]**




$$P(10 \le V \le 14) = \int_{10}^{14} \frac{1}{8}\,dv = \frac{1}{8}(14-10) = \frac{4}{8} = 0.5$$

The shaded area represents half of the support interval.

## 2 Example Linear Density Model **[C2-5]**

Suppose $X$ has density $f(x) = kx$ for $0 \le x \le 2$.

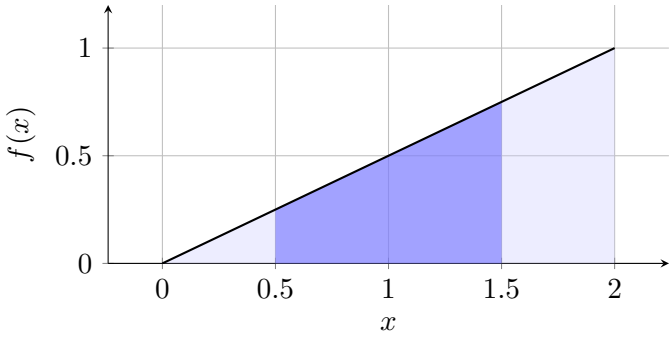
**Questions/Tasks**

**C2** Find the value of $k$. **[3]**

**C2** Compute $P(0.5 \le X \le 1.5)$. **[2]**

**C2 - Find the value of $k$** **[3]**

$$1 = \int_0^2 kx\,dx = k\left[\frac{x^2}{2}\right]_0^2 = 2k \quad \Rightarrow \quad k = \frac{1}{2}.$$




**C2 - Compute $P(0.5 \le X \le 1.5)$** **[2]**

$$P(0.5 \le X \le 1.5) = \int_{0.5}^{1.5} \frac{1}{2}x\,dx = \left[\frac{x^2}{4}\right]_{0.5}^{1.5} = \frac{1.5^2 - 0.5^2}{4} = 0.5$$

## 3 Example Table and Probability Summary [C3-3]

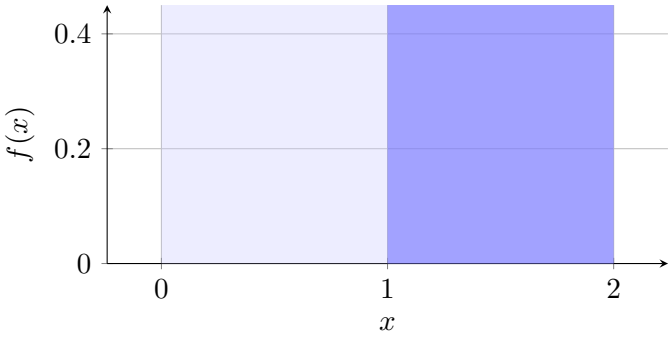
A summary table is shown for two intervals of a piecewise density.

**Questions/Tasks**

**C3** Complete the probability summary table. **[1]**

**C3** State one consistency check. **[2]**

### C3 - Complete the probability summary table [1]




| Interval | Expression | Result |
|---|---|---|
| $[0,1]$ | $\int_0^1 0.4\,dx$ | 0.4 |
| $[1,2]$ | $\int_1^2 0.6\,dx$ | 0.6 |
| Total | $0.4+0.6$ | 1 |

### C3 - State one consistency check [2]

A valid check is that all interval probabilities are nonnegative and add to 1.